

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

B.Tech (Computer Science and Engineering(Artificial Intelligence and Machine Learning)) SEMESTER - 6 Summer 2025 (Regular)

Course :Bachelor of Technology (Computer Science and Engineering(Artificial Intelligence and Machine Learning)) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTAIOE604B - CRYPTOGRAPHY & NETWORK SECURITY

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

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Which of the following is not a principle of security?

- A. Confidentiality B. Availability C. Efficiency D. Integrity

Which attack involves passive listening to communication without altering the message?

- A. Masquerade B. Replay C. Eavesdropping D. Modification

In symmetric key encryption, the same key is used for:

- A. Hashing only B. Encryption and decryption
C. Signature verification only D. Key exchange only

4 Which of the following is NOT an asymmetric key algorithm?

- A. RSA B. Elgamal C. Diffie-Hellman D. RC4

5 Kerberos uses which technique for secure authentication?

- A. One-time pad B. Hash chains
C. Ticket-granting system D. Certificate chaining

In IPsec, which component provides encryption of the payload?

- A. Authentication Header B. ESP
C. Security Association D. Tunnel Mode

Which attack does Cross-Site Scripting (XSS) exploit?

- A. Cryptographic flaw B. Input validation flaw
C. Protocol flaw D. Key exchange flaw

8 Steganography is the technique of:

- A. Encrypting the message using a cipher
B. Hiding the existence of the message
C. Hashing data for authentication
D. Converting plaintext to ciphertext

9 In Elgamal encryption, the decryption requires:

- A. Private key of sender B. Public key of receiver
C. Private key of receiver D. Shared secret key

- 10 In the TLS handshake, the session key is typically established using:
 A. Symmetric key exchange B. One-way hash
 C. Asymmetric key exchange D. Digital watermarking
- 11 In Single Sign-On (SSO), a user logs in:
 A. Once to access one application only B. Multiple times for multiple apps
 C. Once to access multiple systems D. Each time for each page
- 12 CMAC is a type of:
 A. Digital signature B. Message authentication code
 C. Encryption algorithm D. Key distribution protocol
- Q2. Solve the following. 6
- Define the following terms:
 a) Security Mechanism b) Encryption c) Decryption.
 Differentiate between symmetric and asymmetric key cryptography with appropriate diagrams.
- B) Define the following terms: 6
 (a) Plaintext (b) Ciphertext (c) Steganography
 Given a transposition cipher, use the key = 4312 to encrypt the plaintext: 'SECURITY'. Show step-by-step encryption.
- Q3. Solve the following. 6
- Given a system using Diffie-Hellman Key Exchange, Public values: $q = 23$, $\alpha = 2$
 User A selects private key $X_a = 6$
 User B selects private key $X_b = 15$
 Evaluate the shared secret key between A and B. Show calculations.
- List and explain the major features of the AES algorithm. Apply the RSA algorithm to encrypt the message $M = 10$, where $p = 3$, $q = 11$. Compute n , $\phi(n)$, choose $e = 3$, and find the ciphertext C . 6
- Q4. Solve Any Two of the following. 6
- A) Define the following: 6
 (a) Hash Function (b) HMAC (c) CMAC (d) Digital Signature
 (e) Message Authentication Code (MAC) (f) Public Key Infrastructure (PKI)
- Explain the process of message authentication using hash functions. How is it different from encryption? 6
- Apply the HMAC algorithm using the following values and show all steps. 6
 Key = 0b10101010,
 Message = '10110011',
 Hash function = simple XOR of key and message.
- Q5. Solve Any Two of the following. 6
- A) Explain the role and functioning of the SSL/TLS handshake protocol in secure communication. 6
- B) Define the following terms: 6
 a) SSL b) TLS c) HTTPS d) SSH e) 802.11
 f) Handshake Protocol

- C) Explain how the security features of different types of Wi-Fi protections — Open Wi-Fi, WEP, WPA, and WPA2 — are different from each other. Make a table that clearly shows the weaknesses (vulnerabilities) of each one and how the newer versions improved on them. 6

Q6. Solve Any Two of the following.

- A) Define the following terms: 6
 a) Pretty Good Privacy (PGP) b) S/MIME c) Authentication Header (AH)
 List the components of the IPSec architecture and explain their basic roles in securing IP packets.
 Define the following terms: 6
 a) Encapsulating Security Payload (ESP) b) Security Association
 c) Internet Key Exchange (IKE)
 Explain how PGP ensures confidentiality, integrity, and authentication in email communication.
 C) List and briefly explain any three real-world applications of Secure Multiparty Computation (SMC). Explain the working of the Encapsulating Security Payload (ESP) header. What parts of the IP packet does it encrypt and authenticate? 6

*** End ***